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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLEGENCE AND MACHINE LEARNING)

Deep Learning Applications In Medical Image Analysis - Brain Tumor

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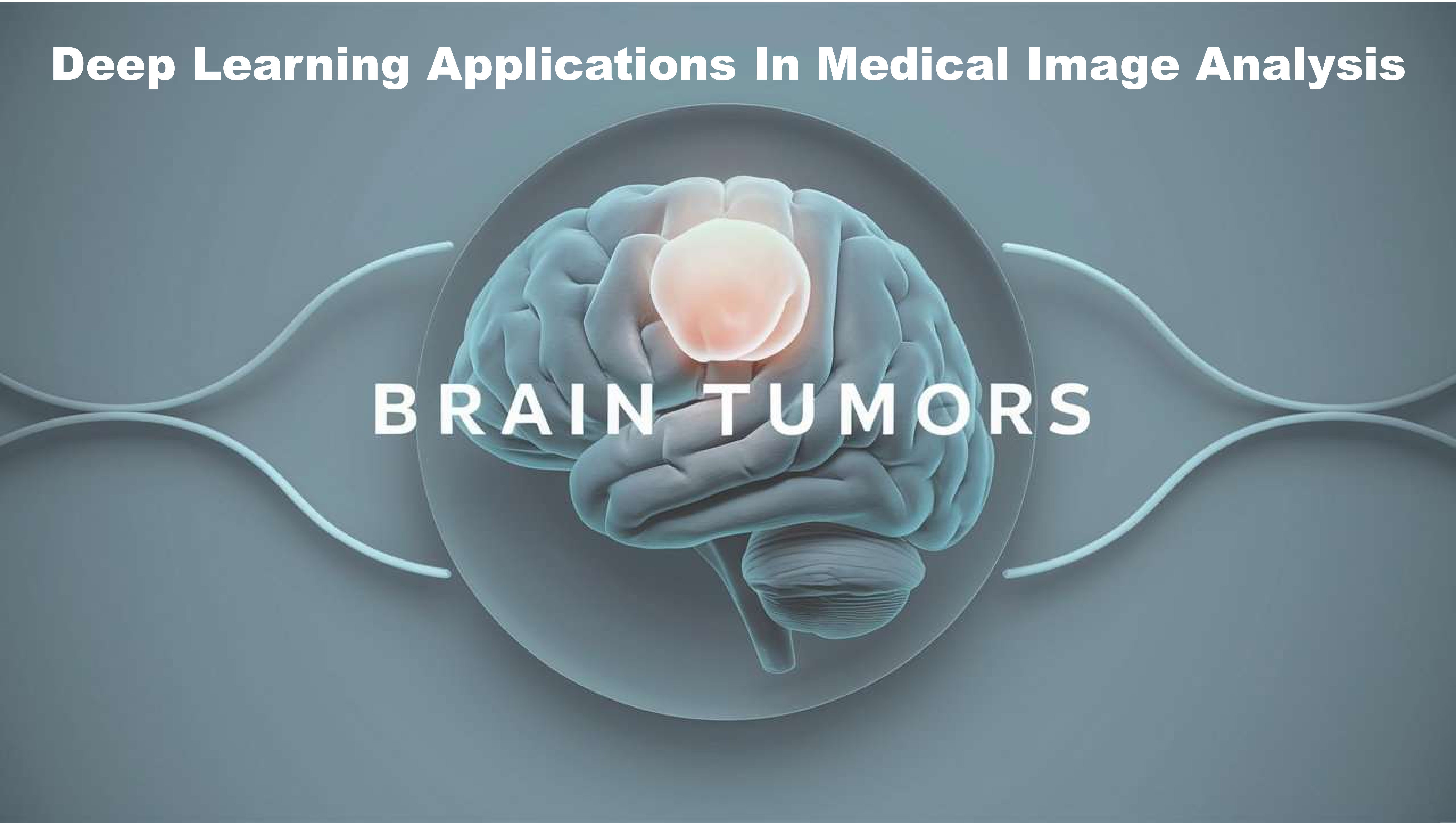

Deep Learning Applications In Medical Image Analysis
BRAIN TUMORS

Agenda

- ABSTRACT
- INTRODUCTION
- EXISTING METHOD
- PROPOSED METHOD
- FLOW CHART
- HARDWARE AND SOFTWARE REQUIREMENTS
- ADVANTAGES
- APPLICATIONS
- RESULT
- CONCLUSION

Abstract

- The project focuses on detecting neurological disorders in the brain using unsupervised machine learning and CNN methods, aiming for high accuracy in diagnosis.
- Early detection of brain diseases like brain tumors and epilepsy is crucial for effective treatment, highlighting the importance of advanced diagnostic techniques.
- The study reviews recent machine learning approaches specifically tailored for identifying these brain diseases, emphasizing their potential in medical diagnostics.
- The integration of unsupervised machine learning and CNN in this project demonstrates a promising advancement in the field of neurology, offering improved methods for early disease detection.

Introduction

- Machine learning algorithms are revolutionizing medicine, from drug discovery to clinical decision-making, as they can analyze vast amounts of digitalized medical data, including electronic health records (EHRs) and medical images.

- The increasing prevalence of EHRs, which include medical images, presents a significant opportunity for machine learning to enhance diagnostic accuracy and efficiency, potentially overcoming limitations faced by human errors.

- Machine learning in medical image analysis improves patient outcomes and helps understand how patients interact with AI-driven healthcare decisions.

Existing Method

- MRI images often have noise and interference, making it hard for doctors to detect tumors early.
- Detecting tumors early is not only difficult, but also takes many days manually.
- These challenges cause problems in the medical field.

Disadvantages

- Detecting tumors manually from noisy MRI images takes a lot of time.
- Early detection of tumors is difficult because of poor image quality.
- These issues lead to delays and errors in diagnosis.

Proposed Method

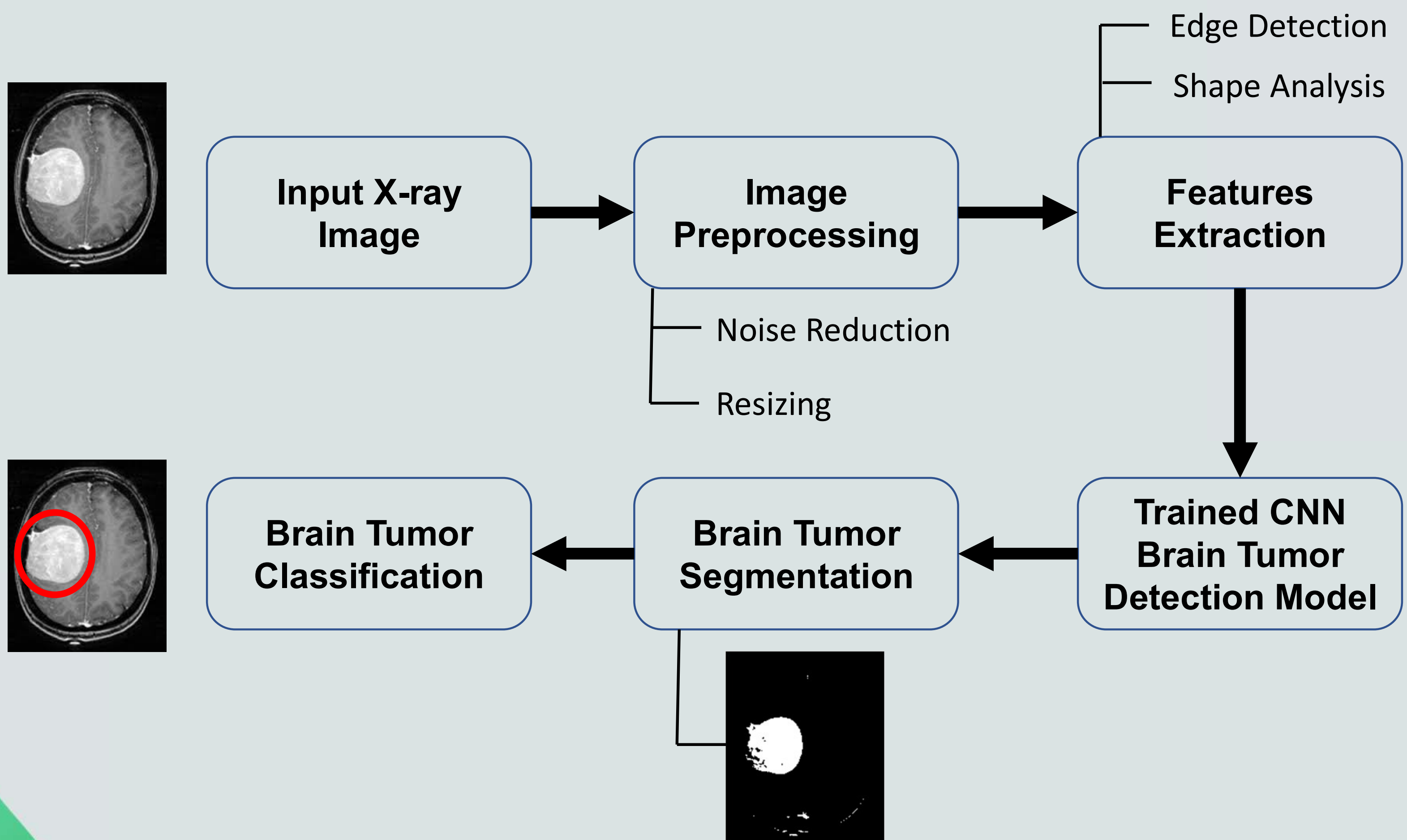
- **Input and Preprocessing**:
  - Uses X-ray/CT scan images of brain tumors and healthy brains, resized to 256 × 256 pixels.
  - Converts images to arrays and applies preprocessing.

- **Data Splitting**:
  - Divides data into 70% training, 20% validation, and 10% testing sets for balanced model training and evaluation.

- **Model Training**:
  - Trains a machine learning model using the preprocessed images to distinguish between brain tumors and healthy brain images.

- **Evaluation and Testing**:
  - Validates the model's performance using the validation set.
  - Tests the final model on the testing set to assess its accuracy and reliability in detecting brain tumors.

Flow Chart


Edge Detection
Shape Analysis
Input X-ray Image
Image Preprocessing
Features Extraction
Noise Reduction
Resizing
Trained CNN Brain Tumor Detection Model
Brain Tumor Segmentation
Brain Tumor Classification

Hardware and Software Requirements

➢Hardware Requirement:

- System Processor : Pentium IV 2.4 GHz or above
- Hard Disk : 20 GB or above
- Monitor : 10/12/15 VGA color
- Ram : 4GB or above

➢Software Requirement:

- Operating system : Windows 7/8/10/11.
- Programming Language : python 3.7
- Type of Application : Graphical User Interface
- Frontend Technologies : Tkinter, CNN with supported libraries
- Backend Technologies : matplotlib, TensorFlow, Keras, etc.
- Dataset : X-ray/CT scan image (Brain)

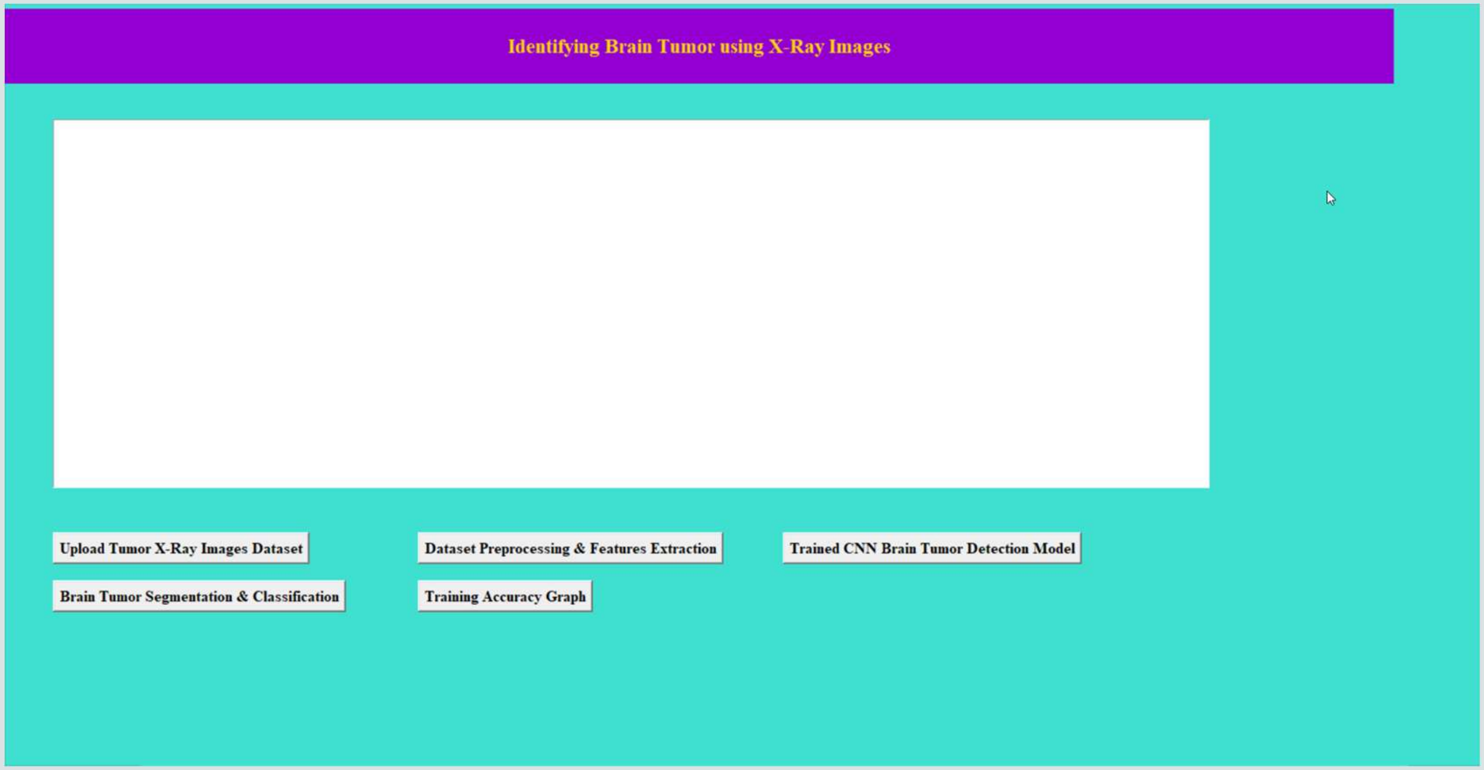
Advantages

- **Better Accuracy**: Detects tumors more accurately than traditional methods.
- **Faster Processing**: Quickly analyzes large amounts of medical images.
- **Consistent Results**: Provides uniform outcomes, reducing human error.
- **Automated**: Reduces the need for manual analysis by doctors.

Applications

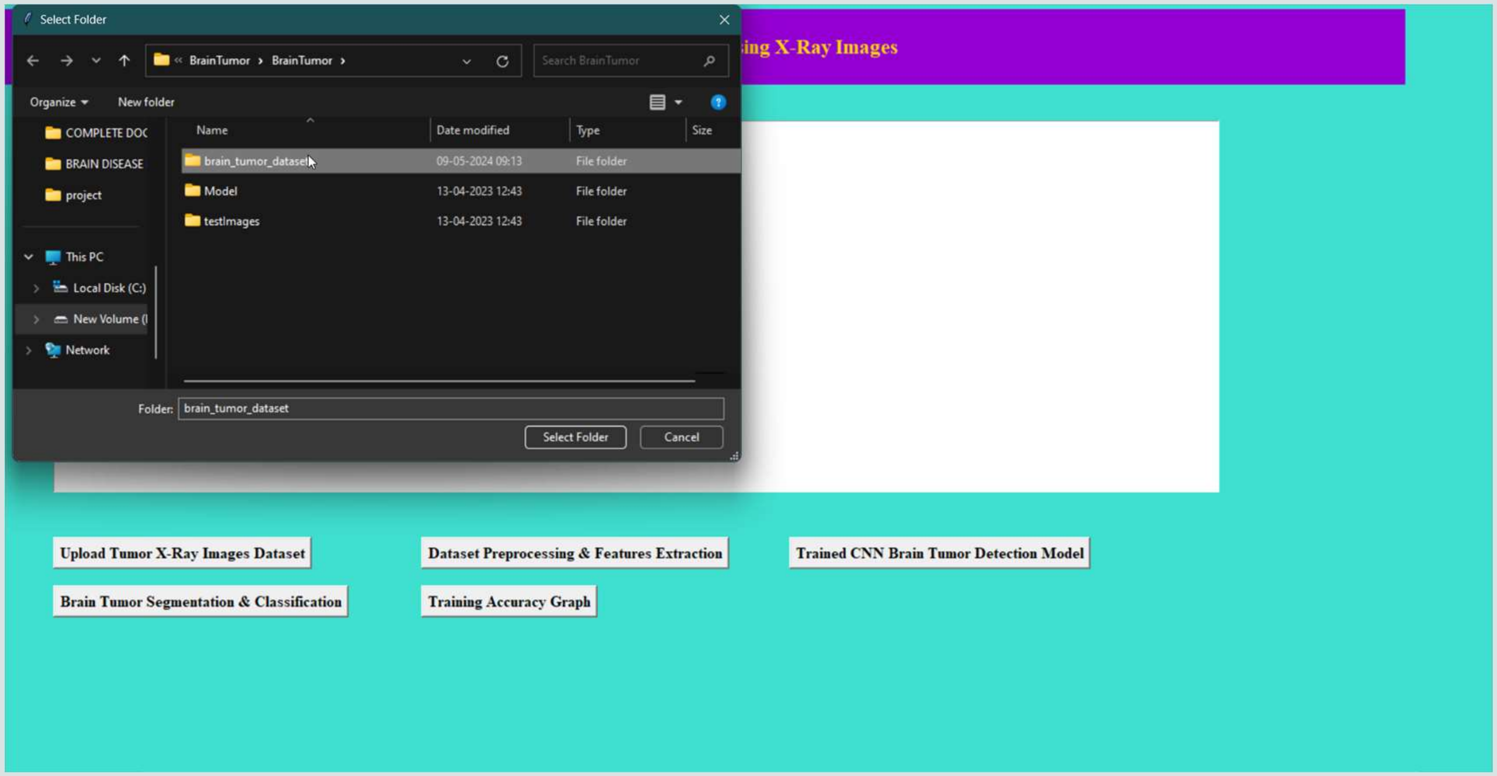
- **Finding Tumors**: Identifies tumors in MRI and CT scans.
- **Classifying Tumors**: Determines the type and grade of tumors.
- **Planning Surgery**: Helps doctors plan where to operate.
- **Tracking Tumors**: Monitors how tumors change over time.

Result



- To upload dataset in above screen click on “upload Tumor X-Ray Images Dataset’ button and get below output.

Result



- In above screen selecting and uploading ‘brain tumor dataset’ dataset folder and then click on ‘Select Folder’ button to upload dataset and to get below output

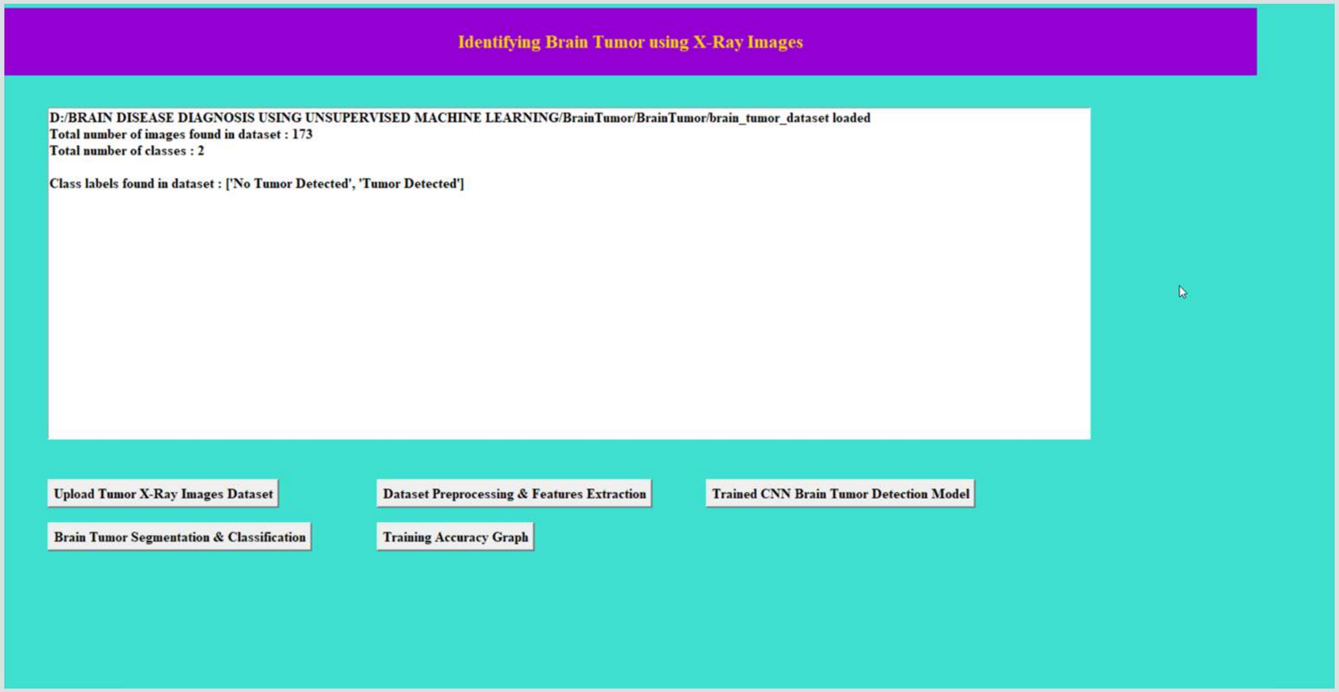
Result

- In above screen dataset loaded and now click on ‘Dataset Preprocessing & Features Extraction’ button to normalize image and get below output.

Result


Identifying Brain Tumor using X-Ray Images
D:/BRAI
NOSIS USING UNSUPERVISED MACHINE LEARNING/BrainTumor/BrainTumor/brain_tumor_dataset loaded
Upload Tumor X-Ray Images Dataset
Dataset Preprocessing & Features Extraction
Trained CNN Brain Tumor Detection Model
Brain Tumor Segmentation & Classification
Training Accuracy Graph

Result

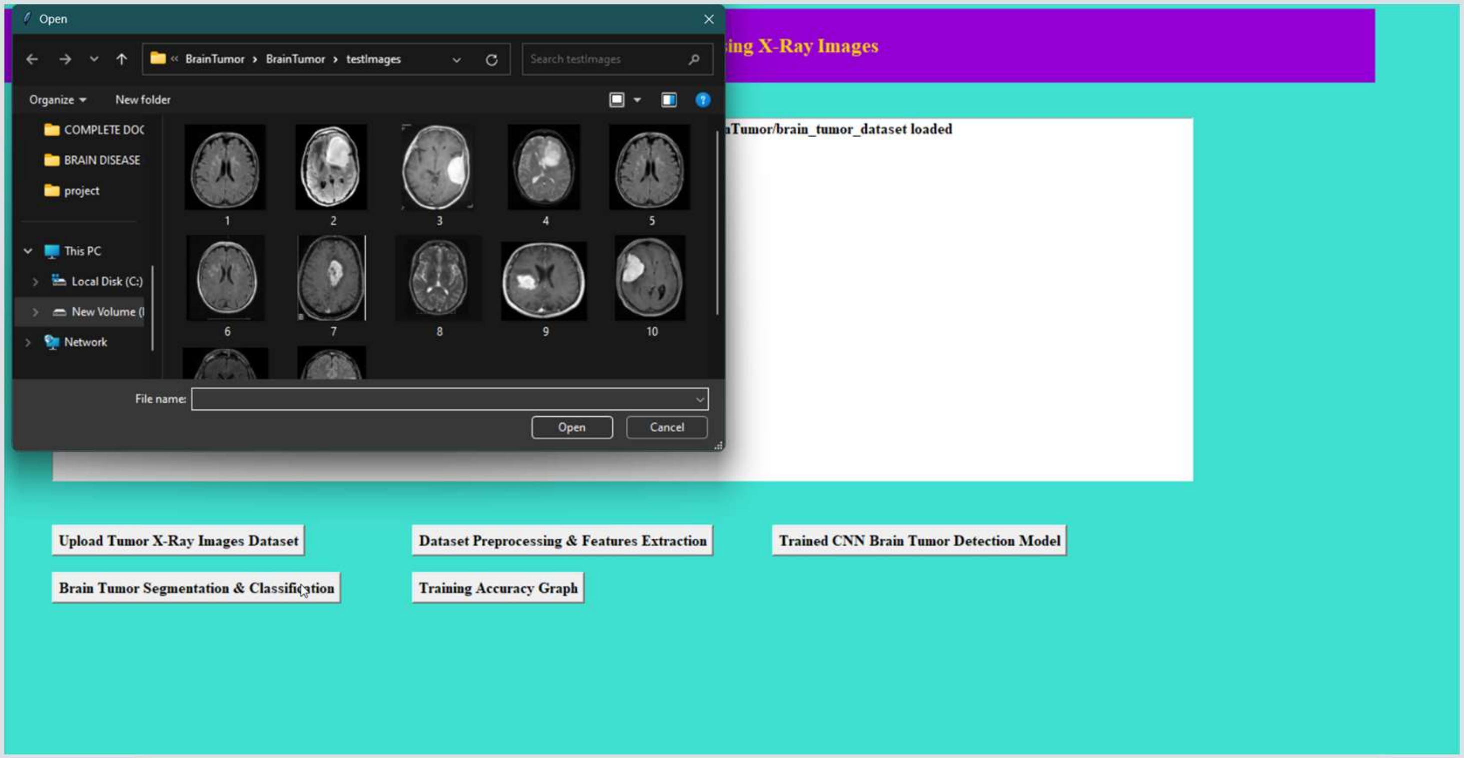


- In above screen all 173 images from dataset are normalized and number of classes are 2 and labels found in dataset are 'No Tumor Detected', 'Tumor Detected' now click the 'Trained CNN Brain Tumor Detection Model' Button to Train and view the Model.

Result

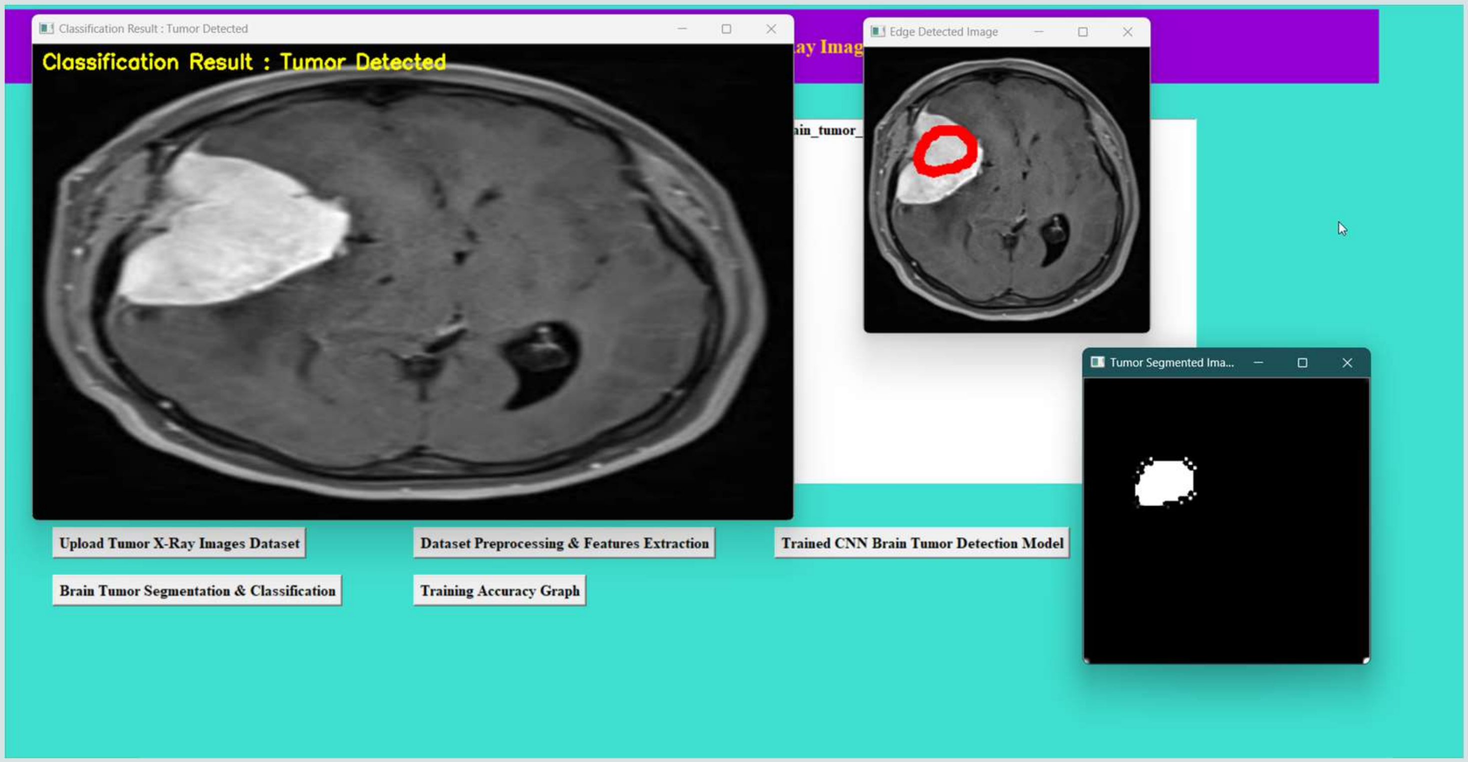
- In above screen we can see that the Brain Tumor Model Generated and the Accuracy on Test Images is 96.06% and Now click on the 'Brain Tumor Segmentation & Classification' Button to Predict the Brain Tumor for New X-ray Image.

Result



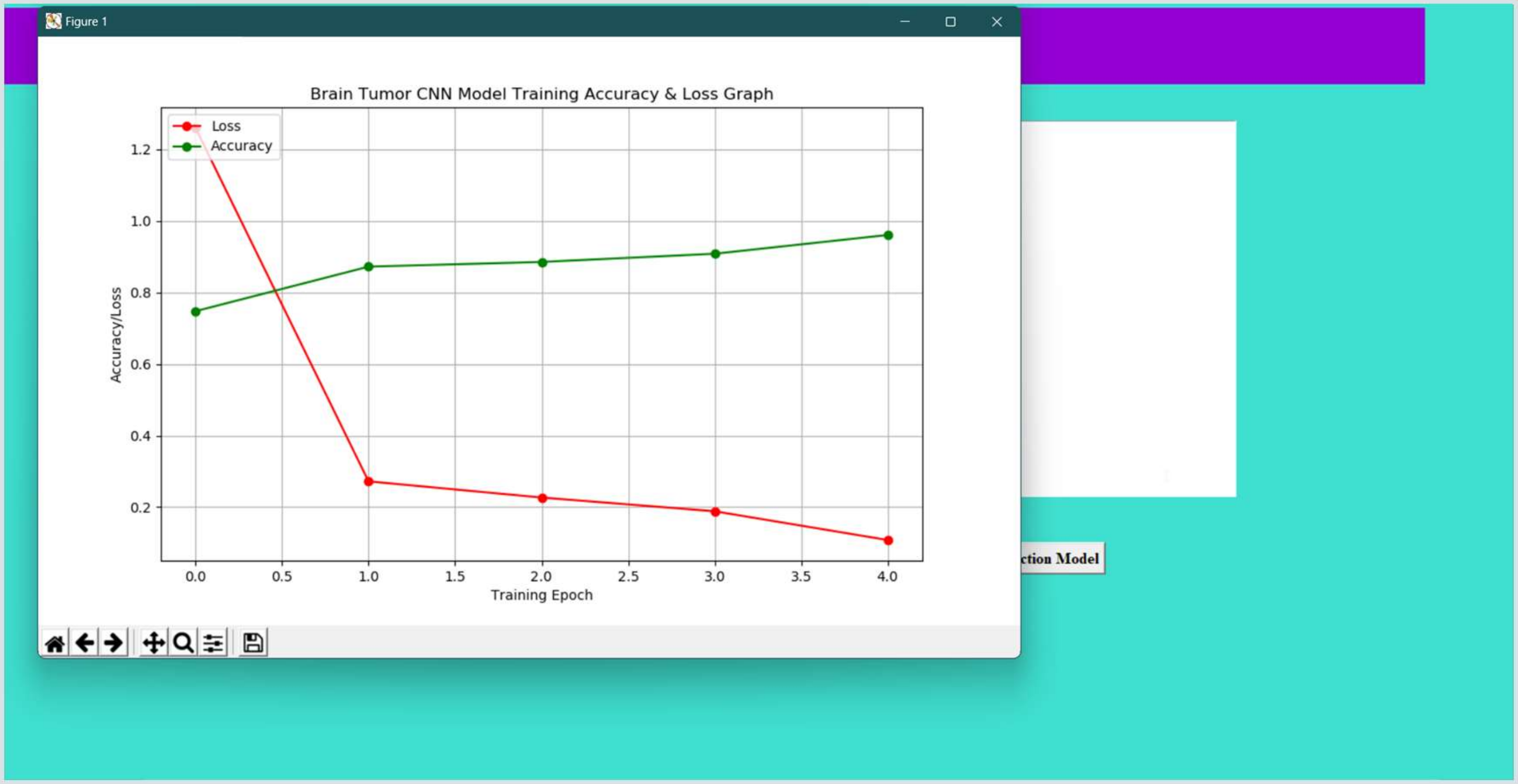
➢ Select the Image to Detect the Brain Tumor and click on Open Now it will generate the result and display the result as below screen.

Result



- This is the result of the Brain Tumor for New X-ray image and it also shows the location of the Brain Tumor with also segmented Image. And next click on 'Training Accuracy Graph' button to view the Graph.

Result



- This is the Graph of Accuracy/Loss vs Training Epoch Graph and shows for every increase in epoch the loss decreases and accuracy increases.

Conclusion

- Improving accuracy in brain disease detection through ML/DL requires more data, hybrid algorithms for clinical integration.
- Challenges include data quality, resource efficiency, and addressing data management, security, and privacy issues.
- For accurate diagnosis of tumor patients, an appropriate segmentation method is required for MRI images to improve diagnosis and treatment.
- Brain Tumor detection aids physicians and benefits the medical imaging industry, especially those producing MRI images.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